Utrecht Summer School

Introduction to Complex Systems

Tuesday projects

Simulating patterns: Cellular automata

1 Patterns & panic (*Python*)

Many spatial patterns observed in complex systems can be explained via evolution rules for each individual element of the system. For example, vegetation patterns in ecosystems can be related to plant interactions; cloud formation can be traced to the evolution of individual droplets, and also social segregation can be viewed in this way.

Pattern formation can occur if the complex system has a spatial scale-dependent feedback mechanism: an element's feature is beneficial for other elements over a short range but detrimental over a longer range. This is also known as 'short-range facilitation' or 'short-range activation' and 'long-range inhibition'. As an example, plants can facilitate further plants growing near them by accumulating moisture and nutrients, while 'stealing' water from further regions where plant growth is then inhibited.

One way to model such spatial interactions is via cellular automata (CA). In this project, we will use a CA simulator that implements a given evolution rule for the individual elements.

1.1 Emergent patterns

Consider a lattice of $M \times N$ cells with periodic boundaries (i.e. the right-side of the domain is connected to the left-side and the bottom-side is connected to the top-side; this is a 'trick' often used to overcome the unwanted effects of having fixed boundaries). All cells can be either passive (0) or active (1). With $A_{ij}(t)$ we denote the value (either 0 or 1) of the site at row i, column j at time t. Now, the evolution rule is based on the spatial scale-dependent feedback. For each cell, we compute the experienced total facilitation/inhibition s_{ij}, where every active state within activation range R_a contributes $+w_a$ and every active state within inhibition range R_i contributes $-w_i$. If $s > 0$, the cell will be active at the next time step; otherwise it will be passive. Mathematically, this evolution rule is given by

$$A_{ij}(t+1) = \begin{cases} 1, & s_{ij}(t) > 0; \\ 0, & \text{otherwise}, \end{cases} \tag{1}$$

where s is given by

$$s_{ij}(t) = \sum_{(k,l)\in N_a(i,j)} A_{kl}(t) w_a - \sum_{(k,l)\in N_i(i,j)} A_{kl}(t) w_i, \tag{2}$$

and $N_a(i,j)$ respectively $N_i(i,j)$ are the collection of cells within activation respectively inhibition range:

$$N_a(i,j) = \{(k,l) : \text{ the distance between } (i,j) \text{ and } (k,l) \text{ is smaller than } R_a\}; \quad (3)$$

$$N_i(i,j) = \{(k,l) : \text{ the distance between } (i,j) \text{ and } (k,l) \text{ is larger than } R_a \text{ and smaller than } R_i\}. \quad (4)$$

To simulate this system, we prepared code for you that provides a `CellularAutomaton` model. Use the Jupyter notebook `pattern_formation.ipynb` to get started. You do not need to modify the source code in `cellular_automaton.py` but you are of course welcome to.

1.1.1 The spatial scale-dependent feedback

The above description depends on the parameters R_a, R_i, w_a and w_i. What kind of conditions do you need to impose on those to allow for pattern formation? Make a diagram for the scale-dependent feedback; i.e. the influence of an 'on' state cell on $s_{ij}(t)$ depending on its distance to the cell (i,j). In the code, the system is modelled on a lattice with periodic boundaries. Does this influence the results? Think about when this is appropriate and when this is not?

To allow for pattern formation neither the facilitation or inhibition should dominate. That means we should have

$$w_a(2R_a)^2 \approx w_i((2R_i)^2 - (2R_a)^2), \quad (5)$$

which can be rewritten as

$$\left(1 + \frac{w_a}{w_i}\right) R_a^2 \approx R_i^2. \quad (6)$$

Periodic boundaries mean that cells are also impacted by those at the other side of the domain. If this wouldn't be the case cells at the edge would have less neighbours to be impacted by. It's a good approximation to model the interior of a large domain without having to specify the boundary conditions, but fails for smaller bounded domains.

1.1.2 Equilibrium states

Run the model for some parameter values that lead to patterns. Will the system converge to an equilibrium state? How does it look like?

Yes, the system will converge to an equilibrium state with some on-state patches in a sea of off.

1.1.3 Dependence on initial conditions

Determine the dependence on the precise initial configuration. For instance, do different patterns emerge when you re-run the code with the same parameter values? Do different patterns emerge for different values of p? (Recall that p denotes the approximate fraction of the initial state that is in the 'on' state.)

For different initial conditions different patterns emerge although the size is roughly similar. The results are independent of the p.

1.1.4 Parameter sensitivity

Experiment with different parameter values and vary the values of R_a, R_i, w_a and w_i. How do the emergent patterns change with these values?

Smaller radii R lead to smaller patterns, i.e. smaller patches of on- and off-state cells. The weights w themselves do not affect the results.

1.1.5 Additional suggestions

If time permits, try to come up with a way to quantify patterns and compare the emergent patterns quantitatively.

Q: Are there standard ways in which this is done? You would want something that tells you something about the average size of the patterns. What is the minimum and maximum distance between two on-cells at the edge of the pattern? How many direct neighbours are in the same state?

1.2 Panic

We now turn our attention to another phenomenon that can be modelled using cellular automata. Let us say that every cell in the lattice now represents a person who can be either panicked (the 'active' state) or unpanicked (the 'passive' state). By stipulating some rule to determine when a person becomes (un)panicked, we can now consider whether and how panic spreads through the system.

We consider the 'droplet rule' for the evolution in this system: for any given person, look at the surrounding individuals (typically taken as the 8 direct neighbouring cells) and count how many of them are panicked. If at least 3 of them are panicked, the person becomes panicked at the next timestep; otherwise the person will be unpanicked.

1.2.1 Implementation

Implement the droplet rule in the code. For this you will need to directly modify the function `CA_evolution`.

HINT: Try to think of the pattern formation system from the first part and think about how this rule is different. Are there activators/inhibitors and what is their range? How large does the strength $s_{ij}(t)$ need to be in order for the person at (i, j) to become panicked? You can implement this with only one change to the evolution function and changing the parameters of the scale-dependent feedback.

With a radius of 2 (1 just means the cell itself) use

$$s_{ij}(t) = \sum_{(k,l)\in N_a(i,j)} A_{kl}(t) - 3 \tag{7}$$

There are no inhibitors and weights are irrelevant. The strength has to be at least 3, i.e. larger than 2. Basically you can set $w_i = 0$, $R_i = 0$, $w_a = 0$ and $R_a = 2$, changing the boundary from 0 to 2

1.2.2 Dependence on initial conditions

Run the model for various values of p. Do you see different behaviour? Why?

The higher p the faster the whole domain is panicked.

1.2.3 Quantification of panic spread

Find a way to quantify the end state of the system (i.e. define some order parameter). Compute its value for various values of p. Does this system have a phase transition? If so, around which p-value does this happen?

When p is small panick dies of quickly. When p is larger it spreads everywhere. Order parameter can be simply counting the fraction of panicked people.

1.2.4 Additional suggestions

If time permits, feel free to experiment with other rules. For example, what happens when you change the number of required panicked neighbours from 3 to another number?

Alternatively, you can also look into the behaviour of your order parameter as the value of p approaches the critical value at which the phase transition happens.

Requiring more cells means a higher p is needed for the panick to spread.